

Software Qualification - RC Laser NA265280

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Notes:

- **All the measurements are made with respect to GND.**
- **Test point numbers and labels are for PCBA Rev F.**

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1. Purpose
2. Explanation of Terminology
3. Equipment required.
4. Laser
 - 4.1 Laser Level
 - 4.2 Laser Plumb1
 - 4.3 Laser Plumb2
 - 4.4 Laser Level Brightness
 - 4.5 Laser Plumb1 Brightness
 - 4.6 Laser Plumb2 Brightness
 - 4.7 Last Laser Turn-On
5. Trigger Buttons
 - 5.1 Level
 - 5.2 Plumb1
 - 5.3 Plumb2
 - 5.4 Brightness
 - 5.5 Translate Up
 - 5.6 Translate Down

Item Tested:	Translate Down Button – Motor Down Movement - Single button press gives out 100 pulses on the oscilloscope. - Motor moves down on press and hold on the button.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	Pass: - The Unit moves Down on the tracks. - There are 100 pulses sent out on test point after single press. Fail: - The unit doesn't move. - There are no pulses sent out on test point after single press.

Test setup:

1. Connect a wire to Test point I55 (net label STEP VERT).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On level laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	101
2	Press button once	101
3	Press button once	101
4	Press button once	101
5	Press button once	101

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	Moving.
2	Button press and hold	Moving.
3	Button press and hold	Moving.
4	Button press and hold	Moving.
5	Button press and hold	Moving.

Table2

Test Outcome

☒ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail

5.8 Counterclockwise (<)

5.9 Power Button

Item Tested:	Power button – Turns the unit On when the Power button is pressed.																																	
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery																																	
Pass / Fail Criteria	Pass: 1. The unit Turn ON when Power button is pressed. 2. The unit Turn OFF when the power button is pressed again. Fail: 1. The Unit doesn't respond / doesn't Turn ON. 2. The unit doesn't repond/ Turnoff on a consecutive button Press.																																	
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. After the insertion on the pack, The RC module does nothing. 3. Press the power button and the Laser diodes and the Leds on FPC are turned ON. 4. Press the power button again and the Laser Diodes and Leds are turned OFF. 5. Record the data in result table below. 6. Repeat the above steps and document the observation. Output: <table><tr><th>Attempt</th><th>Button Action</th><th>State of Unit</th></tr><tr><td>1</td><td>Button press on</td><td>ON</td></tr><tr><td>2</td><td>Button press off</td><td>OFF</td></tr><tr><td>3</td><td>Button press on</td><td>ON</td></tr><tr><td>4</td><td>Button press off</td><td>OFF</td></tr><tr><td>5</td><td>Button press on</td><td>ON</td></tr><tr><td>6</td><td>Button press off</td><td>OFF</td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>				Attempt	Button Action	State of Unit	1	Button press on	ON	2	Button press off	OFF	3	Button press on	ON	4	Button press off	OFF	5	Button press on	ON	6	Button press off	OFF									
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6	Button press off	OFF																																
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail																														

5.10 Pendulum Lock

Item Tested:	Pendulum Lock – Slope led Turn on when pendulum is locked and laser diode have a blink pattern detonating that laser is on slope mode.																														
Item spec:	The laser blink pattern is as follow: • Laser is solid On for 8.5sec • Laser blinks 3 times for next 1.5sec. This 10sec cycles continues as long pendulum is in locked state.																														
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery																														
Pass / Fail Criteria	Pass: 1. The slope LED turn On when the pendulum lock is in locked position. 2. The laser diodes blink pattern is as explained in the spec. Fail: 1. The slope LED does not turn On when the pendulum lock is in locked position. 2. The laser diodes blink pattern is out of spec.																														
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Slide the pendulum lock to locked position. 3. Press the power button. 4. Turn off any laser that are ON. 5. Turn the level laser on. 6. Toggle pendulum lock to unlocked position and back to locked position. 7. Record the blink pattern on the level laser. 8. Change the brightness and repeat steps 12 and 13. 9. Record the data in result table below. Output: <table><tr><th>Attempt</th><th>Brightness (High/med/low)</th><th>State of blink pattern</th></tr><tr><td>1</td><td>High</td><td>ON/Blinking as per test.</td></tr><tr><td>2</td><td>Med</td><td>ON/Blinking as per test.</td></tr><tr><td>3</td><td>Low</td><td>ON/Blinking as per test.</td></tr><tr><td>4</td><td>High</td><td>ON/Blinking as per test.</td></tr><tr><td>5</td><td>Med</td><td>ON/Blinking as per test.</td></tr><tr><td>6</td><td>Low</td><td>ON/Blinking as per test.</td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	Attempt	Brightness (High/med/low)	State of blink pattern	1	High	ON/Blinking as per test.	2	Med	ON/Blinking as per test.	3	Low	ON/Blinking as per test.	4	High	ON/Blinking as per test.	5	Med	ON/Blinking as per test.	6	Low	ON/Blinking as per test.									
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5	Med	ON/Blinking as per test.																													
6	Low	ON/Blinking as per test.																													

Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail

6. Status LEDs

6.1 BLE LED

Item Tested:	Battery SOC – LED illumination for Bluetooth status
Equipment needed:	RC Laser module flashed and programmed, Programmable power supply, Dewalt battery pack power supply adapter
Pass / Fail Criteria	Pass: - The unit displays the correct Bluetooth status Fail: - ANY time the Bluetooth led does not indicate the correct status.

Test Procedure:

1. Insert the Dewalt battery pack into the laser.
2. Turn the Unit ON by pressing the Power button
3. Observe the status of BLE led.
4. Ble Led blink at 1Hz for 60 sec.
5. Ble LED becomes solid ON when a remote is connected to it.
6. Record the observation.

Output:

NOTE: LED status options are: ON, OFF, BLINK

Sn.no	BLE Led status
1 Start Blinking.	OFF after 60 sec.
2 Start Blinking.	OFF after 60 sec.
3 Start Blinking.	OFF after 60 sec.
4 Start Blinking.	OFF after 60 sec.
5 Start Blinking.	OFF after 60 sec.
6 Start Blinking.	OFF after 60 sec.
7 Start Blinking.	OFF after 60 sec.
8 Start Blinking.	OFF after 60 sec.

Test Outcome

☐ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail

6.2 Battery SOC

Item Tested:

Battery SOC – LED illumination versus battery voltage

Equipment needed:	RC Laser module flashed and programmed, Programmable power supply, Dewalt battery pack power supply adapter																																																																											
Pass / Fail Criteria	Pass: 2. The unit displays the correct SOC LEDs based on battery voltage Fail: 2. ANY time the SOC LEDs do not indicate the correct SOC level																																																																											
Test Procedure:	7. Using the battery pack to power supply adapter, connect the 25V power supply to the RC Laser. 8. Set the power supply current limit to 300mA +/- 50mA. 9. Set the power supply to the first voltage in the table. 10. Record the status of the SOC LEDs in the table. 11. Set the power supply to the second voltage in the table. 12. Record the status of the SOC LEDs in the table. 13. Continue to set the voltages and record the status of the SOC LEDs until all voltages in the table have been tested 14. Repeat the above steps 3 times and document the observation. Output: NOTE: LED status options are: ON, OFF, BLINK <table border="1"> <thead> <tr> <th>Power supply voltage</th> <th>LED 1 (top)</th> <th>LED 2 (middle)</th> <th>LED 3 (bottom)</th> </tr> </thead> <tbody> <tr><td>18.3V +/- 0.05V</td><td>ON</td><td>ON</td><td>ON</td></tr> <tr><td>18.1V +/- 0.05V</td><td>OFF</td><td>ON</td><td>ON</td></tr> <tr><td>16.6V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>ON</td></tr> <tr><td>16.4V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>ON</td></tr> <tr><td>14.9V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>Blink</td></tr> <tr><td>14.7V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>Blink</td></tr> <tr><td>14.1V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>OFF</td></tr> <tr><td>13.9V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>OFF</td></tr> <tr><td colspan="4"> </td></tr> <tr><td>13.9V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>OFF</td></tr> <tr><td>14.1V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>OFF</td></tr> <tr><td>14.7V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>Blink</td></tr> <tr><td>14.9V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>Blink</td></tr> <tr><td>16.4V +/- 0.05V</td><td>OFF</td><td>OFF</td><td>ON</td></tr> <tr><td>16.6V +/- 0.05V</td><td>OFF</td><td>ON</td><td>Blink</td></tr> <tr><td>18.1V +/- 0.05V</td><td>OFF</td><td>ON</td><td>Blink</td></tr> <tr><td>18.3V +/- 0.05V</td><td>ON</td><td>ON</td><td>Blink</td></tr> </tbody> </table>				Power supply voltage	LED 1 (top)	LED 2 (middle)	LED 3 (bottom)	18.3V +/- 0.05V	ON	ON	ON	18.1V +/- 0.05V	OFF	ON	ON	16.6V +/- 0.05V	OFF	OFF	ON	16.4V +/- 0.05V	OFF	OFF	ON	14.9V +/- 0.05V	OFF	OFF	Blink	14.7V +/- 0.05V	OFF	OFF	Blink	14.1V +/- 0.05V	OFF	OFF	OFF	13.9V +/- 0.05V	OFF	OFF	OFF					13.9V +/- 0.05V	OFF	OFF	OFF	14.1V +/- 0.05V	OFF	OFF	OFF	14.7V +/- 0.05V	OFF	OFF	Blink	14.9V +/- 0.05V	OFF	OFF	Blink	16.4V +/- 0.05V	OFF	OFF	ON	16.6V +/- 0.05V	OFF	ON	Blink	18.1V +/- 0.05V	OFF	ON	Blink	18.3V +/- 0.05V	ON	ON	Blink
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Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail																																																																								

6.3 Slope LED

Item Tested:	Slope LED : Slope LED turn on when the pendulum is moved to lock position.																																	
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery																																	
Pass / Fail Criteria	Pass: 3. The slope Turn ON when Pendulum is moved locked position. Fail: 1. Slope LED doesn't Turn ON.																																	
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Press the power button and turn off any laser diode that are On. 3. Slide the position of pendulum lock switch to locked position. 4. Record the state of Slope LED in result table below. 5. Slide the position of pendulum lock switch to unlocked position. 6. Repeat the above steps and document the observation. Output: <table border="1"> <thead> <tr> <th>Attempt</th> <th>State of Pendulum</th> <th>State of LED</th> </tr> </thead> <tbody> <tr><td>1</td><td>Locked</td><td>ON</td></tr> <tr><td>2</td><td>Unlocked</td><td>OFF</td></tr> <tr><td>3</td><td>Locked</td><td>ON</td></tr> <tr><td>4</td><td>Unlocked</td><td>OFF</td></tr> <tr><td>5</td><td>Locked</td><td>ON</td></tr> <tr><td>6</td><td>Unlocked</td><td>OFF</td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </tbody> </table>				Attempt	State of Pendulum	State of LED	1	Locked	ON	2	Unlocked	OFF	3	Locked	ON	4	Unlocked	OFF	5	Locked	ON	6	Unlocked	OFF									
Attempt	State of Pendulum	State of LED																																
1	Locked	ON																																
2	Unlocked	OFF																																
3	Locked	ON																																
4	Unlocked	OFF																																
5	Locked	ON																																
6	Unlocked	OFF																																
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail																														

6.4 Drop LED

Item Tested:	Drop LED – Drop LED turn on when the unit is dropped																	
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery																	
Pass / Fail Criteria	Pass: - The unit Drop LED turns ON when Unit is dropped more than 1m. Fail: - The Drop LED doesn't turn ON when the unit is dropped.																	
Test Procedure:	- Insert the Dewalt battery pack to RC Laser. - Press the power button again and Turn off any laser that are ON. - Drop the Unit on the test bench and record the state of drop led. - Repeat the above steps and document the observation. Output: <table border="1"> <thead> <tr> <th>Attempt</th> <th>State of LED</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>ON</td> </tr> <tr> <td>2</td> <td>ON</td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> </tbody> </table>				Attempt	State of LED	1	ON	2	ON								
Attempt	State of LED																	
1	ON																	
2	ON																	
Test Outcome	☐ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail														

7. BLE connection

7.1 BLE New remote connection

Item Tested:	BLE LED on Pairing new remote
Equipment needed:	RC Laser and Remote

Pass / Fail Criteria	Pass: - On pressing Bluetooth button on remote after 60 sec of turning ON the Laser Unit, the BLE connection LED on laser unit is OFF. - On pressing Bluetooth button on remote within 60 sec of turning on the laser the BLE connection LED on laser unit STAYS ON. Fail: - On pressing Bluetooth button on remote after 60 sec of turning ON the Laser Unit, the BLE connection LED on laser unit is not OFF. - On pressing Bluetooth button on remote within 60 sec of turning on the laser the BLE connection LED on laser unit does not STAY ON.																										
Test Setup:	TEST SETUP: - Laser unit - Remote - Timer/Stopwatch																										
Test Procedure:	- Turn on remote by inserting the battery. - Turn on the laser and timer for 60 sec. - Press and hold the Bluetooth button on remote after 60 sec. - Check BLE led. - Press and hold the Bluetooth button on remote to unpair the remote. - Turn off the laser and then turn on the laser again. - Press Bluetooth button on remote before 60 sec. - Check BLE led. - Repeat the above step and record observation. Output: <table><tr><th>S.no</th><th>Timer</th><th>State of LED</th></tr><tr><td rowspan="2">1</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr><tr><td rowspan="2">2</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr><tr><td rowspan="2">3</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr><tr><td rowspan="2">4</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr></table>				S.no	Timer	State of LED	1	>60 Sec	OFF	<60sec	ON	2	>60 Sec	OFF	<60sec	ON	3	>60 Sec	OFF	<60sec	ON	4	>60 Sec	OFF	<60sec	ON
S.no	Timer	State of LED																									
1	>60 Sec	OFF																									
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2	>60 Sec	OFF																									
	<60sec	ON																									
3	>60 Sec	OFF																									
	<60sec	ON																									
4	>60 Sec	OFF																									
	<60sec	ON																									
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail																							

7.2 Connection to Paired Remote

Item Tested:	BLE LED turn ON connecting to already paired remote																										
Equipment needed:	RC Laser and Remote																										
Pass / Fail Criteria	Pass: - When laser is turned on with the presence of already paired remote, the blue led (BLE LED) turns ON (solid, not blinking) irrespective of 60 sec time. Fail: - When laser is turned on with the presence of already paired remote, the blue led (BLE LED) does not turn ON (solid, not blinking) irrespective of 60 sec time.																										
Test Setup:	TEST SETUP: - Laser unit - Remote - Timer/Stopwatch																										
Test Procedure:	- Do this after “pair new remote” test - Turn on the laser and timer for 60 sec. - After 60 sec, turn on remote by inserting the battery. - Check BLE led. - Turn off the laser and then turn on the laser again (within 60 sec). - Check BLE led. - Repeat the above step and record observation. Output: <table><tr><th>S.no</th><th>Timer</th><th>State of LED</th></tr><tr><td rowspan="2">1</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr><tr><td rowspan="2">2</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr><tr><td rowspan="2">3</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr><tr><td rowspan="2">4</td><td>>60 Sec</td><td>OFF</td></tr><tr><td><60sec</td><td>ON</td></tr></table>				S.no	Timer	State of LED	1	>60 Sec	OFF	<60sec	ON	2	>60 Sec	OFF	<60sec	ON	3	>60 Sec	OFF	<60sec	ON	4	>60 Sec	OFF	<60sec	ON
S.no	Timer	State of LED																									
1	>60 Sec	OFF																									
	<60sec	ON																									
2	>60 Sec	OFF																									
	<60sec	ON																									
3	>60 Sec	OFF																									
	<60sec	ON																									
4	>60 Sec	OFF																									
	<60sec	ON																									
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail																							

7.3 Unpair a Paired Remote

Item Tested:	BLE LED turn off when an Already paired remote LRC is Unpaired.																	
Equipment needed:	RC Laser and Remote																	
Pass / Fail Criteria	Pass: 1. When an already Paired Remote is Unpaired from the Unit, the BLE LED turn Off indicating a successful unpair. 2. After Unpair the Remote cannot control the Laser Fail: 1. When an already Paired Remote is Unpaired from the Unit, the BLE LED doesn't turn Off. 2. After Unpair the Remote can still control the Laser																	
Test Setup:	TEST SETUP: 1. Laser Unit, Remote, Battery pack, timer.																	
Test Procedure:	1. Do this after "pair new remote" test 2. Turn on the laser and timer for 60 sec. 3. After 60 sec, turn on remote by inserting the battery. 4. Check BLE led. 5. Press And Hold the BLE Button on the LRC to unpair the remote. 6. Check BLE led. 7. Repair the remote and follow above steps again and note the observation. Output: <table><tr><th>Attempt</th><th>State of LED</th></tr><tr><td>1</td><td>OFF</td></tr><tr><td>2</td><td>OFF</td></tr><tr><td>3</td><td>OFF</td></tr><tr><td>4</td><td>OFF</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>				Attempt	State of LED	1	OFF	2	OFF	3	OFF	4	OFF				
Attempt	State of LED																	
1	OFF																	
2	OFF																	
3	OFF																	
4	OFF																	
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail														

8. Safety Timeout

8.1 30 sec motor Timeout

Item Tested:	Timeout of vertical and yaw motors												
Item Spec:	Motor must stop running after 30 seconds of continuous rotation												
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, Vertical motor, Yaw motor, Timer or stopwatch												
Pass / Fail Criteria	Pass: 1. The vertical motor stops spinning after 30 seconds +0.5s of continuous rotation 2. The yaw motor stops spinning after 30 seconds +0.5s of continuous rotation Fail: 1. The vertical and/or yaw motor continues to spin after 30 seconds of continuous rotation.												
Test Setup:	Test SETUP: 1. Connect the vertical motor to the vertical motor connector (J5). 2. Connect the yaw motor to the yaw motor connector (J6). 3. Insert the Dewalt battery into the RC Laser unit.												
Test Procedure:	Test PROCEDURE: 1. Reset the timer. 2. Press and hold the down button on the laser unit. Start the timer as soon as the button is pressed. 3. Record the elapsed time that the vertical motor is spinning. Release the button when the vertical motor stops spinning or after 35 seconds. 4. Reset the timer. 5. Press and hold the up button on the laser unit. Start the timer as soon as the button is pressed. 6. Record the elapsed time that the vertical motor is spinning. Release the button when the vertical motor stops spinning or after 35 seconds. 7. Reset the timer. 8. Press and hold the '<' button on the laser unit. Start the timer as soon as the button is pressed. 9. Record the elapsed time that the yaw motor is spinning. Release the button when the yaw motor stops spinning or after 35 seconds. 10. Reset the timer. 11. Press and hold the '>' button on the laser unit. Start the timer as soon as the button is pressed. 12. Record the elapsed time that the yaw motor is spinning. Release the button when the yaw motor stops spinning or after 35 seconds. Output: <table border="1"> <thead> <tr> <th>Button press</th><th>Elapsed time</th></tr> </thead> <tbody> <tr> <td>Down button</td><td>Stop</td></tr> <tr> <td>Up button</td><td>Stop</td></tr> <tr> <td>'<' button</td><td>Stop</td></tr> <tr> <td>'>' button</td><td>Stop</td></tr> </tbody> </table>			Button press	Elapsed time	Down button	Stop	Up button	Stop	'<' button	Stop	'>' button	Stop
Button press	Elapsed time												
Down button	Stop												
Up button	Stop												
'<' button	Stop												
'>' button	Stop												
	☒ Pass	☐ Fail	☐ Conditional Pass ☐ Pass after Fail										

Test Outcome				
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8.2 BLE Disconnect Motor Timeout

Item Tested:	BLE Disconnect Stop the any motor that running.
Item Spec:	Motor must stop running as soon as remote disconnects.
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, Vertical motor, Yaw motor, Timer or stopwatch
Pass / Fail Criteria	Pass: 1. The Motor that is running, stop as soon as remote disconnects. Fail: 1. The Motor doesn't stop running after the remote disconnects.
Test Setup:	Test SETUP: 1. Connect the vertical motor to the vertical motor connector (J5). 2. Connect the yaw motor to the yaw motor connector (J6). 3. Insert the Dewalt battery into the RC Laser unit.

Test Procedure:

Test PROCEDURE:

1. Insert the Battery pack in the laser Unit.
2. Press the power button to turn the unit On.
3. Press any button on the LRC so it connects to the Laser unit.
4. Press and hold Up Button to run the vertical motor continuously.
5. Within 30sec, and while holding the up button, remove the Battery from the remote so it disconnects.
6. Observe the state of the motor and record the observation in the table below.
7. Repeat the above with different motor buttons to activate both the motors in all directions.

Output:

Sno.	Button press	State of motor after remote disconnect
1	Up button	Stop moving
2		
3	Down Button	Stop moving
4		
5	'<' button	Stop moving
6		
7	'>' button	Stop moving
8		

Test Outcome

☒ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail**8.3 Thermistor Cutoff**

Item Tested:	Hot pack shutdown – The unit shuts down when the Temp > 70C
Item Spec:	Unit should shut down when the temperature reached > 70C.
Equipment needed:	RC Laser module flashed and programmed, Power supply, Resistance box.
Pass / Fail Criteria	Pass: 1. The Turns off, when the temp reaches > 70C(resistance between B+ and TH reaches 2.2k). Fail: 1. The unit doesn't turn off when temp > 70C.

Test Setup:	Test SETUP: - Set the Power supply to 20V@300mA. - Set the resistance box to 10k Ohms. - Connect the input of resistance box to B+ and output to TH on Laser unit.																										
Test Procedure:	Test PROCEDURE: - Connect B+ and GND on Laser unit to the power supply. - Connect the TH terminal on the Laser Unit to the resistance box set at 10k Ohms. - Press the power button to turn the unit On. - Reduce the Resistance to 2.2k Ohms. - Observe the state of the Unit and record the observation in the table below. Output: <table border="1"> <thead> <tr> <th>S.no</th> <th>Resistance</th> <th>State of the Unit</th> </tr> </thead> <tbody> <tr> <td rowspan="2">1</td> <td>10</td> <td>ON</td> </tr> <tr> <td>2.2</td> <td>OFF</td> </tr> <tr> <td rowspan="2">2</td> <td>10</td> <td>ON</td> </tr> <tr> <td>2.2</td> <td>OFF</td> </tr> <tr> <td rowspan="2">3</td> <td>10</td> <td>ON</td> </tr> <tr> <td>2.2</td> <td>OFF</td> </tr> <tr> <td rowspan="2">4</td> <td>10</td> <td>ON</td> </tr> <tr> <td>2.2</td> <td>OFF</td> </tr> </tbody> </table>				S.no	Resistance	State of the Unit	1	10	ON	2.2	OFF	2	10	ON	2.2	OFF	3	10	ON	2.2	OFF	4	10	ON	2.2	OFF
S.no	Resistance	State of the Unit																									
1	10	ON																									
	2.2	OFF																									
2	10	ON																									
	2.2	OFF																									
3	10	ON																									
	2.2	OFF																									
4	10	ON																									
	2.2	OFF																									
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail																							

8.4 Low Voltage Cutoff

Item Tested:	Low Voltage Cutoff
Item Spec:	Unit should shutdown when the voltage reached below 14.1V
Equipment needed:	RC Laser module flashed and programmed, Power supply,

Pass / Fail Criteria	Pass: - The Turns off, when the temp reaches < 14.1V Fail: - The unit doesn't turn off when temp < 14.1V.																																																						
Test Setup:	Test SETUP: Set the Power supply to 20V@300mA.																																																						
Test Procedure:	Test PROCEDURE: - Connect B+ and GND on Laser unit to the power supply. - Press the Power button to turn the Unit ON. - Reduce the voltage down to 14.5V - Reduce the voltage down to 13.8V in steps of 0.1V and record the observation when the unit cuts off. - Repeat the above step 3 times. Output: <table border="1"> <thead> <tr> <th>S.no</th><th>Volatge</th><th>State of the Unit</th></tr> </thead> <tbody> <tr><td rowspan="8">1</td><td>14.5</td><td>ON</td></tr> <tr><td>14.4</td><td>ON</td></tr> <tr><td>14.3</td><td>ON</td></tr> <tr><td>14.2</td><td>ON</td></tr> <tr><td>14.1</td><td>OFF</td></tr> <tr><td>14</td><td>OFF</td></tr> <tr><td>13.9</td><td>OFF</td></tr> <tr><td>13.8</td><td>OFF</td></tr> <tr><td rowspan="8">2</td><td>14.5</td><td>ON</td></tr> <tr><td>14.4</td><td>ON</td></tr> <tr><td>14.3</td><td>ON</td></tr> <tr><td>14.2</td><td>ON</td></tr> <tr><td>14.1</td><td>OFF</td></tr> <tr><td>14</td><td>OFF</td></tr> <tr><td>13.9</td><td>OFF</td></tr> <tr><td>13.8</td><td>OFF</td></tr> <tr><td rowspan="8">3</td><td>14.5</td><td>ON</td></tr> <tr><td>14.4</td><td>ON</td></tr> <tr><td>14.3</td><td>ON</td></tr> <tr><td>14.2</td><td>ON</td></tr> <tr><td>14.1</td><td>OFF</td></tr> <tr><td>14</td><td>OFF</td></tr> <tr><td>13.9</td><td>OFF</td></tr> <tr><td>13.8</td><td>OFF</td></tr> </tbody> </table>	S.no	Volatge	State of the Unit	1	14.5	ON	14.4	ON	14.3	ON	14.2	ON	14.1	OFF	14	OFF	13.9	OFF	13.8	OFF	2	14.5	ON	14.4	ON	14.3	ON	14.2	ON	14.1	OFF	14	OFF	13.9	OFF	13.8	OFF	3	14.5	ON	14.4	ON	14.3	ON	14.2	ON	14.1	OFF	14	OFF	13.9	OFF	13.8	OFF
S.no	Volatge	State of the Unit																																																					
1	14.5	ON																																																					
	14.4	ON																																																					
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	14	OFF																																																					
	13.9	OFF																																																					
	13.8	OFF																																																					
2	14.5	ON																																																					
	14.4	ON																																																					
	14.3	ON																																																					
	14.2	ON																																																					
	14.1	OFF																																																					
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	13.8	OFF																																																					
3	14.5	ON																																																					
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	14.1	OFF																																																					
	14	OFF																																																					
	13.9	OFF																																																					
	13.8	OFF																																																					
Test Outcome	☒ Pass ☐ Fail ☐ Conditional Pass ☐ Pass after Fail																																																						

8.5 Laser Over current Protection Cutoff

Item Tested:	Laser Over current protection cutoff																		
Item Spec:	Lasers should turnoff when the current reaches above set threshold for each of laser diodes.																		
Equipment needed:	RC Laser module flashed and programmed, Power supply,																		
Pass / Fail Criteria	Pass: 1. The laser turns off when the current increases beyond the threshold acc to table below. <table border="1"> <thead> <tr> <th>Laser diode</th><th>Current cutoff</th></tr> </thead> <tbody> <tr> <td>Level</td><td>350mA</td></tr> <tr> <td>Plumb1</td><td>300mA</td></tr> <tr> <td>Plumb2</td><td>300mA</td></tr> <tr> <td>Level+Plumb1</td><td>650mA</td></tr> <tr> <td>Level+Plumb2</td><td>650mA</td></tr> <tr> <td>Plumb1 + Plumb2</td><td>600mA</td></tr> <tr> <td>Level+plumb1+plumb2</td><td>950mA</td></tr> <tr> <td></td><td></td></tr> </tbody> </table> Fail: 1. The laser do not turn off above a reached threshold.	Laser diode	Current cutoff	Level	350mA	Plumb1	300mA	Plumb2	300mA	Level+Plumb1	650mA	Level+Plumb2	650mA	Plumb1 + Plumb2	600mA	Level+plumb1+plumb2	950mA		
Laser diode	Current cutoff																		
Level	350mA																		
Plumb1	300mA																		
Plumb2	300mA																		
Level+Plumb1	650mA																		
Level+Plumb2	650mA																		
Plumb1 + Plumb2	600mA																		
Level+plumb1+plumb2	950mA																		
Test Setup:	Test SETUP: DO NOT EXCEED VOLATGE >0.9V ON Test Point I136 1. Set a Power supply at 20V@300mA and connect it to B+ and GND on the terminal. 2. Connect a wire between Test point I136(NET LABEL LASER CURRENT) and another power supply. 3. Connect a wire to Test point I124 (Net label EN_P2) 4. Connect a wire to Test point I128 (Net label EN_L) 5. Connect a wire to Test point I132 (Net label EN_P1)																		

Test Procedure:

Test PROCEDURE:

1. Press the Power button to turn the Unit ON.
2. Disable the power supply at I136.
3. Press the Level Laser button and verify it is on by the signal at test point I128. It should have a PWM signal.
4. Set the power supply connected to I136 to 0.30V and enable the power supply.
5. Increase the voltage on power supply in step of 0.01V until level turns off verified by signal on Test point I128.
6. Disable the power supply connected to I136.
7. Press the Plumb1 Laser button and verify it is on by the signal at test point I132. It should have a PWM signal.
8. Set the power supply connected to I136 to 0.25V and enable the power supply.
9. Increase the voltage on power supply in step of 0.01V until level turns off verified by signal on Test point I136.
10. Disable the power supply connected to I136.
11. Press the Plumb2 Laser button and verify it is on by the signal at test point I124. It should have a PWM signal.
12. Set the power supply connected to I136 to 0.25V and enable the power supply.
13. Increase the voltage on power supply in step of 0.01V until level turns off verified by signal on Test point I124.
14. Disable the power supply connected to I136.
15. Press the Level laser button and the Plumb1 laser button. Verify they are both on by the signals at I128 and I132. They should both have PWM signals.
16. Set the power supply connected to I136 to 0.60V and enable the power supply.
17. Increase the voltage on power supply in step of 0.01V until level and plumb1 turns off verified by signal on Test point I128 and I132.
18. Disable the power supply connected to I136.
19. Press the Level laser and Plumb2 laser buttons. Verify they are both on by the signals at I128 and I124. They should both have PWM signals.
20. Set the power supply connected to I136 to 0.60V and enable the power supply.
21. Increase the voltage on power supply in step of 0.01V until Level and Plumb2 turns off verified by signal on Test point I128 and I132.
22. Disable the power supply connected to I136.
23. Press the Plumb1 laser and Plumb2 laser buttons. Verify they are both on by the signals at I124 and I132. They should both have PWM signals.
24. Set the power supply connected to I136 to 0.55V and enable the power supply.
25. Increase the voltage on power supply in step of 0.01V until Plumb1 and Plumb2 turns off verified by signal on Test point I124 and I132.
26. Disable the power supply connected to I136.
27. Press the Level laser, Plumb1 laser, and Plumb2 laser buttons. Verify they are all on by the signals at I124, I128, and I132. They should all have PWM signals.
28. Set the power supply connected to I136 to 0.90V and enable the power supply.
29. Increase the voltage on step of 0.01V until Level laser, Plumb1 laser, and Plumb2 laser are all off verified by the signals at test points I124, I128, and I132. They should all be <0.2V.
30. Disable the power supply connected to I136.
31. Disable the power supply connected to I92 (Net label B+)
32. Repeat the above step 3 times.

Output:

Trial no	Lasers tested	Voltage at I136	Level laser status (on/off)	Plumb1 laser status (on/off)	Plumb2 laser status (on/off)
1	L	0.30V	ON	N/A	N/A
		0.31V	ON		
		0.32V	ON		
		0.33V	ON		
		0.34V	OFF		
		0.35V			
		0.36V			
		0.37V			
		0.38V			
	P1	0.25V	N/A	ON ON ON ON OFF	N/A
		0.26V			
		0.27V			
		0.28V			
		0.29V			
		0.30V			
		0.31V			
		0.32V			
	P2	0.25V	N/A	N/A	ON ON ON ON OFF
		0.26V			
		0.27V			
		0.28V			
		0.29V			
		0.30V			
		0.31V			
		0.32V			
	L, P1	0.60V	ON ON OFF	ON ON OFF	N/A
		0.61V			
		0.62V			
		0.63V			
		0.64V			
		0.65V			
		0.66V			
		0.67V			
	L, P2	0.6V	ON ON OFF	N/A	ON ON OFF
		0.61V			
		0.62V			
		0.63V			
		0.64V			
		0.65V			
		0.66V			
		0.67V			
	P1, P2	0.55V	N/A	ON ON OFF	ON ON OFF
		0.56V			
		0.57V			
		0.58V			
		0.59V			

	2	L, P1, P2	0.60V	ON OFF	ON OFF	ON OFF
			0.61V			
			0.62V			
			0.9V			
			0.91V			
			0.92V			
			0.93V			
			0.94V			
			0.95V			
			0.96V			
			0.97V			
		L	0.30V	ON ON ON ON OFF	N/A	N/A
			0.31V			
			0.32V			
			0.33V			
			0.34V			
			0.35V			
			0.36V			
			0.37V			
			0.38V			
		P1	0.25V	N/A	ON ON ON ON OFF	N/A
			0.26V			
			0.27V			
			0.28V			
			0.29V			
			0.30V			
			0.31V			
			0.32V			
		P2	0.25V	N/A	N/A	ON ON ON ON OFF
			0.26V			
			0.27V			
			0.28V			
			0.29V			
			0.30V			
			0.31V			
			0.32V			
		L, P1	0.60V	ON ON OFF	ON ON OFF	N/A
			0.61V			
			0.62V			
			0.63V			
			0.64V			
			0.65V			
			0.66V			
			0.67V			
		L, P2	0.6V	ON ON OFF	N/A	ON ON OFF
			0.61V			
			0.62V			
			0.63V			

			0.64V			
			0.65V			
			0.66V			
			0.67V			
		P1, P2	0.55V	N/A	ON ON OFF	ON ON OFF
			0.56V			
			0.57V			
			0.58V			
			0.59V			
			0.60V			
			0.61V			
			0.62V			
		L, P1, P2	0.9V	ON OFF	ON OFF	ON OFF
			0.91V			
			0.92V			
			0.93V			
			0.94V			
			0.95V			
			0.96V			
			0.97V			
	3	L	0.30V	ON ON ON ON OFF	N/A	N/A
			0.31V			
			0.32V			
			0.33V			
			0.34V			
			0.35V			
			0.36V			
			0.37V			
			0.38V			
		P1	0.25V	N/A	ON ON ON ON OFF	N/A
			0.26V			
			0.27V			
			0.28V			
			0.29V			
			0.30V			
			0.31V			
			0.32V			
		P2	0.25V	N/A	N/A	ON ON ON ON OFF
			0.26V			
			0.27V			
			0.28V			
			0.29V			
			0.30V			
			0.31V			
			0.32V			
		L, P1	0.60V	ON ON OFF	ON ON OFF	N/A
			0.61V			
			0.62V			

			0.63V					
			0.64V					
			0.65V					
			0.66V					
			0.67V					
	L, P2		0.6V	ON ON OFF	N/A	ON ON OFF		
			0.61V					
			0.62V					
			0.63V					
			0.64V					
			0.65V					
			0.66V					
			0.67V					
	P1, P2		0.55V	N/A	ON ON OFF	ON ON OFF		
			0.56V					
			0.57V					
			0.58V					
			0.59V					
			0.60V					
			0.61V					
			0.62V					
	L, P1, P2		0.9V	ON OFF	ON OFF	ON OFF		
			0.91V					
			0.92V					
			0.93V					
			0.94V					
			0.95V					
			0.96V					
			0.97V					
Test Outcome	☒ Pass		☐ Fail		☐ Conditional Pass		☐ Pass after Fail	

9. Accelerometer

9.1 Out of level blink

Item Tested:	Laser diode blink are out of level is detected
Item Spec:	All the laser diodes ON must blink at 1Hz if unit is out of level only if the unit is calibrated
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, Vertical motor, Yaw motor, Timer or stopwatch, remote control

Pass / Fail Criteria	Pass: 1. The laser diode blink at 1Hz as soon as Unit is tilted more than 4 degrees. 2. The laser diode blink at 1Hz as soon as the Unit is overturned. 3. The laser diodes do Not blink for the above points if the unit is not calibrated. Fail: 1. The laser diodes do not blink and stay solid on when unit is tilted. 2. The laser diode blink at 1Hz as soon as the Unit is overturned. 3. The laser diodes blink if the unit is not calibrated.																																																																														
Test Setup:	Test SETUP: 1. Connect the Laser head cable(white cable) in the J7 connector on the main unit board. 2. Connect the Dewalt battery. 3. Remote connection is not necessary for this test.																																																																														
Test Procedure:	Test PROCEDURE: 1. Make sure that the unit is not calibrated. 2. Turn the unit ON by pressing the Power On button 3. Slide pendulum to unlocked position. 4. By default Laser Level should turn ON. 5. Turn ON the plumb1 and plumb 2 laser ON. 6. Tilt the unit to > 4 degree with battery as a pivot point on the xaxis. 7. Record the observation 8. Return the unit back to 0 degree and record the observation 9. Tilt the unit to > 4 degree with battery as a pivot point on the yaxis. 10. Record the Observation. 11. Overturn the unit where unit becomes Upside down 12. Record the observation 13. Return the unit back to 0 degree where the unit is resting on the battery. 14. Repeat the above steps 3 times. 15. Complete section 12(calibration test) and repeat the entire procedure again and record the observation . Output: Unit not calibrated: <table><tr><th>X axis</th><th>Blink(yes/no)</th><th>Y axis</th><th>Blink(yes/no)</th><th>Z axis</th><th>Blink(yes/no)</th></tr><tr><td>4</td><td>NO</td><td>4</td><td>NO</td><td>Overturned</td><td>NO</td></tr><tr><td>0</td><td>NO</td><td>0</td><td>NO</td><td>0</td><td>NO</td></tr><tr><td>4</td><td>NO</td><td>4</td><td>NO</td><td>Overturned</td><td>NO</td></tr><tr><td>0</td><td>NO</td><td>0</td><td>NO</td><td>0</td><td>NO</td></tr><tr><td>4</td><td>NO</td><td>4</td><td>NO</td><td>Overturned</td><td>NO</td></tr><tr><td>0</td><td>NO</td><td>0</td><td>NO</td><td>0</td><td>NO</td></tr></table> Unit is calibrated: <table><tr><th>X axis</th><th>Blink(yes/no)</th><th>Y axis</th><th>Blink(yes/no)</th><th>Z axis</th><th>Blink(yes/no)</th></tr><tr><td>4</td><td>Yes</td><td>4</td><td>Yes</td><td>Overturned</td><td>Yes</td></tr><tr><td>0</td><td>No</td><td>0</td><td>No</td><td>0</td><td>No</td></tr><tr><td>4</td><td>Yes</td><td>4</td><td>Yes</td><td>Overturned</td><td>Yes</td></tr><tr><td>0</td><td>No</td><td>0</td><td>No</td><td>0</td><td>No</td></tr><tr><td>4</td><td>Yes</td><td>4</td><td>Yes</td><td>Overturned</td><td>Yes</td></tr></table>	X axis	Blink(yes/no)	Y axis	Blink(yes/no)	Z axis	Blink(yes/no)	4	NO	4	NO	Overturned	NO	0	NO	0	NO	0	NO	4	NO	4	NO	Overturned	NO	0	NO	0	NO	0	NO	4	NO	4	NO	Overturned	NO	0	NO	0	NO	0	NO	X axis	Blink(yes/no)	Y axis	Blink(yes/no)	Z axis	Blink(yes/no)	4	Yes	4	Yes	Overturned	Yes	0	No	0	No	0	No	4	Yes	4	Yes	Overturned	Yes	0	No	0	No	0	No	4	Yes	4	Yes	Overturned	Yes
X axis	Blink(yes/no)	Y axis	Blink(yes/no)	Z axis	Blink(yes/no)																																																																										
4	NO	4	NO	Overturned	NO																																																																										
0	NO	0	NO	0	NO																																																																										
4	NO	4	NO	Overturned	NO																																																																										
0	NO	0	NO	0	NO																																																																										
4	NO	4	NO	Overturned	NO																																																																										
0	NO	0	NO	0	NO																																																																										
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4	Yes	4	Yes	Overturned	Yes																																																																										
0	No	0	No	0	No																																																																										
4	Yes	4	Yes	Overturned	Yes																																																																										
0	No	0	No	0	No																																																																										
4	Yes	4	Yes	Overturned	Yes																																																																										

	0	No	0		0	
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail		

10. Laser control through Remote

10.1 Level

Item Tested:	Level Laser operation by remote
Item Spec:	Level Laser turns ON and OFF using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote
Pass / Fail Criteria	Pass: 2. Level Laser turn ON or OFF using the remote control Fail: 3. Level laser doesn't turn ON when it is off and the Level Laser button is pressed. 4. Level laser doesn't turn OFF when it is on and the Level Laser button is pressed.

Test Procedure:

1. Insert the 20V DeWalt battery pack into the laser unit.
2. Slide the pendulum lock to unlocked position.
3. Press the power button on the laser unit.
4. Turn off any lasers that are on.
5. Press the Level Laser Button on the remote.
6. Record the data in the result table below.
7. Press the Level Laser Button on the remote.
8. Record the data in the result table below.
9. Repeat steps 5-8 two more times (total 6 button presses).

Output:

Press number	Level laser status (ON/OFF)
1	ON/OFF
2	ON/OFF
3	ON/OFF
4	ON/OFF
5	ON/OFF
6	ON/OFF

Test Outcome

☒ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail

10.2 Plumb1

Item Tested:

Plumb1 Laser operation by remote

Item Spec:

Plumb1 Laser turns ON and OFF using the remote control

Equipment needed:

RC Laser module flashed and programmed, Dewalt Battery, remote

Pass / Fail Criteria

Pass:

3. Plumb1 Laser turn ON or OFF using the remote control

Fail:

5. Plumb1 laser doesn't turn ON when it is off and the Plumb1 Laser button is pressed.
6. Plumb1 laser doesn't turn OFF when it is on and the Plumb1 Laser button is pressed.

Test Procedure:

10. Insert the 20V DeWalt battery pack into the laser unit.
11. Slide the pendulum lock to unlocked position.
12. Press the power button on the laser unit.
13. Turn off any lasers that are on.

14. Press the Plumb1 Laser Button on the remote.
15. Record the data in the result table below.
16. Press the Plumb1 Laser Button on the remote.
17. Record the data in the result table below.
18. Repeat steps 5-8 two more times (total 6 button presses).

Output:

Press number	Plumb1 Laser status (ON/OFF)
1	ON/OFF
2	ON/OFF
3	ON/OFF
4	ON/OFF
5	ON/OFF
6	ON/OFF

Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail
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10.3 Plumb2

Item Tested:	Plumb2 Laser operation by remote
Item Spec:	Plumb2 Laser turns ON and OFF using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote
Pass / Fail Criteria	Pass: 4. Plumb2 Laser turn ON or OFF using the remote control Fail: 7. Plumb2 laser doesn't turn ON when it is off and the Plumb2 Laser button is pressed. 8. Plumb2 laser doesn't turn OFF when it is on and the Plumb2 Laser button is pressed.
Test Procedure:	19. Insert the 20V DeWalt battery pack into the laser unit. 20. Slide the pendulum lock to unlocked position. 21. Press the power button on the laser unit. 22. Turn off any lasers that are on. 23. Press the Plumb2 Laser Button on the remote. 24. Record the data in the result table below. 25. Press the Plumb2 Laser Button on the remote. 26. Record the data in the result table below. 27. Repeat steps 5-8 two more times (total 6 button presses).

	Output:			
	Press number	Level laser status (ON/OFF)		
	1	ON/OFF		
	2	ON/OFF		
	3	ON/OFF		
	4	ON/OFF		
	5	ON/OFF		
	6	ON/OFF		
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail

10.4 Translate Up

Item Tested:	Translate Up operation by remote - Single button press gives out 100 pulses on oscilloscope. - Motor moves up on press and hold on the button.
Item Spec:	Unit moves UP using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote
Pass / Fail Criteria	Pass: - The Unit moves Up on the tracks. - There are 100 pulses sent out on test point after single press. Fail: - The unit doesn't move. - There are no pulses sent out on test point after single press.
Test Setup:	- Connect a wire to Test point I55 (net label STEP VERT). - Connect this to oscilloscope to measure freq. - Set up the oscilloscope. - Connect RC Laser terminal to battery pack.
Test Procedure:	- Insert the Dewalt battery pack to RC Laser. - Slide the pendulum lock to unlocked position. - Turn On level laser.

4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	102
2	Press button once	102
3	Press button once	102
4	Press button once	102
5	Press button once	102

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	Moving
1	Button press and hold	Moving
2	Button press and hold	Moving
3	Button press and hold	Moving
4	Button press and hold	Moving

Table2

Test Outcome

☒ Pass

☐ Fail

☐ Conditional Pass

☐ Pass after Fail

10.5 Translate Down

Item Tested:	Translate Down operation by remote • Single button press gives out 100 pulses on oscilloscope. • Motor moves up on press and hold on the button.
Item Spec:	Unit moves down using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote

Pass / Fail Criteria

Pass:

1. The Unit moves Down on the tracks.
2. There are 100 pulses sent out on test point after single press.

Fail:

1. The unit doesn't move.
2. There are no pulses sent out on test point after single press.

Test Setup:

1. Connect a wire to Test point I55 (net label STEP VERT).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On level laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	102
2	Press button once	102
3	Press button once	102
4	Press button once	102
5	Press button once	102

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	Moving
2	Button press and hold	Moving
3	Button press and hold	Moving
4	Button press and hold	Moving
5	Button press and hold	Moving

Table2

Test Outcome

☒ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail

10.6 Clockwise(>)

Item Tested:

Translate Clockwise Operation by Remote

- Single button press gives out 4 pulses on the oscilloscope.
- Motor moves clockwise on press and hold on the button.

Equipment needed:

RC Laser module flashed and programmed, Dewalt Battery

Pass / Fail Criteria

Pass:

1. The Unit moves clockwise on the tracks.
2. There are 4 pulses sent out on test point after single press.

Fail:

1. The unit doesn't move.
3. There are no pulses sent out on test point after single press.

Test setup:

1. Connect a wire to Test point I51 (net label STEP YAW).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On plumb1 laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	04

2	Press button once	04
3	Press button once	04
4	Press button once	04
5	Press button once	04

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	Moving
2	Button press and hold	Moving
3	Button press and hold	Moving
4	Button press and hold	Moving
5	Button press and hold	Moving

Table2

Test Outcome

☒ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail

10.7 Counterclockwise (<)

Item Tested:

Translate Counter Clockwise operation by remote

- Single button press gives out 4 pulses on the oscilloscope.
- Motor moves clockwise on press and hold on the button.

Equipment needed:

RC Laser module flashed and programmed, Dewalt Battery

Pass / Fail Criteria

Pass:

1. The Unit moves clockwise on the tracks.
2. There are 4 pulses sent out on test point after single press.

Fail:

1. The unit doesn't move.
4. There are no pulses sent out on test point after single press.

Test setup:

1. Connect a wire to Test point I51 (net label STEP YAW).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On plumb1 laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	04
2	Press button once	04
3	Press button once	04
4	Press button once	04
5	Press button once	04

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	Moving
2	Button press and hold	Moving
3	Button press and hold	Moving
4	Button press and hold	Moving
5	Button press and hold	Moving

Table2

Test Outcome

☒ Pass☐ Fail☐ Conditional Pass☐ Pass after Fail

Item Tested:	Datalogging parameters.			
Item Spec:	Datalogging works with the SBD Ibox and the parameters are changing per runtime.			
Equipment needed:	RC Laser module flashed and programmed, Dewalt batter, SBD IBox.			
Pass / Fail Criteria	Pass: - Parameters change per run using when read through Ibox. Fail: - Parameter do not change per run when read through IBox.			
Test Procedure:	Test PROCEDURE: - Insert the DeWalt pack to Ibox. - Connect the IBox to the computer and start the bootloader app. - Insert the Ibox in the laser unit. - Press the power button to turn the unit ON. - Read the Datalog information from the bootloader app. - Note the value of the switch cycle and runtime. - Remove the ibox from the unit, insert the pack and press the Laser level , plumb1 button. - Follow steps from 3 to 6 and record the observation in the output section. - Share the snippet from the bootloader section. Output:  esl2-3 datalogging.xlsx			
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail

12. Calibration

Item Tested:	Calibration using the IBox.
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, SBD IBox.

Pass / Fail Criteria	Pass: 1. Calibration complete										
	Fail: 1. Calibration incomplete										
Test Procedure:	Test PROCEDURE: 1. Insert the DeWalt pack to Ibox. 2. Connect the IBox to the computer and start the bootloader app. 3. Insert the Ibox in the laser unit. 4. Press the power button to turn the unit ON. 5. In the Bootloader app, press the Calibrate button as shown in the image above. 6. Notice the state of the unit. 7. Level laser turn on indicating the start of calibration process. 8. After some time, plumb/vertical laser turn on Indicating the end of Calibration. 9. After some time, all the laser turn off indicating the calibration routine has been completed. 10. Repeat the above steps 3 times and record in table below										
	Output: <table border="1"> <thead> <tr> <th>Sno</th> <th>Unit Calibrated</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>Yes</td> </tr> <tr> <td>1</td> <td>YES</td> </tr> <tr> <td>2</td> <td>YES</td> </tr> </tbody> </table>				Sno	Unit Calibrated	1	Yes	1	YES	2
Sno	Unit Calibrated										
1	Yes										
1	YES										
2	YES										
Test Outcome	☒ Pass	☐ Fail	☐ Conditional Pass	☐ Pass after Fail							

Document revision history

Rev	Description	Author	Date	Reviewed
1.0	Initial document Software qualification document	Kavya Jain	21 June 2023	N/A
1.1	Added Equipment needed	Kenn Norr	22 June 2023	
2.0	Added the Motor Timeout	Kenn Norr		
3.0	ESL1 Software qual for Bangalore	Kavya Jain		
4.0	ESL3 Software qual, Added 8.3, 8.4, 8.5	Kavya Jain	19 Feb 2024	
4.1	Added Remote operation and modified 8.5	Kenn	20 Feb	
5.0	Changed the test point of all the tests and added calibration and datalogging	Kavya	21	
5.1	Added Drop LED and Slop LED	Kavya	22	
5.2	Added sec 9 Out of level blink	Kavya	25	
6.0	added single press test for Motor operation. Sec 5.5, 5.6, 5.7, 5.8	Kavya	26	

6.1	Modified sec 8.5 and correct Cutoff and	Kenn	26	
6.2	Added BLE unpair remote test	Kavya	28	
7.0	Updated changes/correction requested by Bangalore. • Updated 5.5 and 5.6 from 4 to 100 pulses • Updated 5.7 and 5.8 to 4 pulses. • Modified procedure for 8.5 • Added no calibration check to sec 9 • Changed the name to v7.0	Kavya	01 march	
7.1	Updated the test points with correct vales on 4.2 nad 4.3	Kavya	04 April	
7.2	Updated 10.4, 10.5, 10.6, 10.7	Kavya	15 April	